

Local goodness-of-fit measures for neural decoding

Sirui Zeng¹, Alison E. Comrie⁵, Loren M. Frank^{2,3,4},

Uri T. Eden¹, Eric L. Denovellis^{2,4,*}

¹Department of Mathematics and Statistics, Boston University, Boston, MA, USA

²Departments of Physiology and Psychiatry, University of California, San Francisco, San Francisco, CA, USA

³Kavli Institute for Fundamental Neuroscience, University of California, San Francisco, San Francisco, CA, USA

⁴Howard Hughes Medical Institute, University of California, San Francisco, San Francisco, CA, USA

⁵Howard Hughes Medical Institute, Janelia Research Campus, Ashburn, VA, USA

*Corresponding author. Email: eric.denovellis@ucsf.edu

ORCID iDs: Sirui Zeng [0009-0001-4056-552X](https://orcid.org/0009-0001-4056-552X); Alison E. Comrie [0000-0002-2251-4892](https://orcid.org/0000-0002-2251-4892); Loren M. Frank [0000-0002-1752-5677](https://orcid.org/0000-0002-1752-5677); Uri T. Eden [0000-0002-2058-3691](https://orcid.org/0000-0002-2058-3691); Eric L. Denovellis [0000-0003-4606-087X](https://orcid.org/0000-0003-4606-087X).

Abstract

State-space models are widely used in neuroscience to infer latent states from observations, but validating them remains difficult. Interpretable models are necessarily simplified relative to the data, and determining the level of complexity needed to describe latent variables accurately is challenging. This challenge results from two main factors. First, because latent variables are inherently unobservable, decoding quality cannot be validated directly against observed data. Second, prediction errors caused by model misspecification can be difficult to interpret. We therefore developed metrics designed to identify localized periods when individual observations conflict with the model's accumulated estimate. Each metric assesses the consistency between the

information contained in individual observations and the information extracted by the state-space model from a local window of data. The metrics fall into three categories: 1) measures of overlap between estimated credible regions obtained from the state distributions; 2) predictive checks for individual observations, based on samples drawn from the assumed state-space model; and 3) the Kullback–Leibler (KL) divergence, included as a familiar baseline that quantifies difference rather than consistency. We applied these metrics to simulated data and to real rat hippocampal neural data, demonstrating their ability to identify periods of model misspecification and to provide interpretable feedback for model improvement. Together, the metrics provide complementary tools for neuroscientists to assess the local goodness-of-fit of state-space models, enabling more informed decisions about model refinement and interpretation of neural decoding results.

Keywords: Neural decoding, State-space models, Goodness-of-fit

Introduction

Internal brain states related to memory, emotion, and attention play central roles in behavior. They arise from a time-varying interplay between the external world and internal processes and are not directly observable, making them difficult to characterize.

Recordings of neural activity, including spike trains, calcium signals, and voltage traces, provide experimental measurements influenced by these latent states. State space models make it possible to relate these measurements to estimates of the latent states themselves. Specifically, these models provide a method for relating complex, high-dimensional neural activity to estimates of low-dimensional latent states, and have provided insights into neural representations and dynamics associated with taste [1], spatial location [2], and aggression [3].

State space models aggregate information about the latent states over time. They combine the information from the data at the current time (the normalized likelihood distribution) with the accumulated information used to predict the latent state (the prediction distribution). This powerful combination enables robust estimation of the latent state, effectively balancing current information

with the accumulated history of the system and prior knowledge.

Despite the wide use of state space models in neuroscience, methods to validate these models remain underdeveloped and underused. One reason is that latent states are inherently difficult to evaluate [4]. In the absence of ground truth, model validity can only be inferred indirectly, such as by comparing relative likelihoods under multiple models, the quality of reconstructed signals, or performance on downstream tasks. Neural data are also dynamic, so model fit may vary substantially over time. Compounding this problem is that common methods of validating models, such as cross-validation and “shuffle” permutation tests, are holistic evaluations of the model — they tell neuroscientists whether or not the model generalizes well or is different from a simplified model with less utility — but they fail to indicate when the model fits poorly or which components are responsible. Consequently, these methods can reject models that are intentionally simplified yet scientifically meaningful and offer little guidance for targeted improvement of the model. For example, analyses to decode rat movements from hippocampal place cells typically ignore known features of the rat’s movement (such as momentum) and known features of neurons (such as refractoriness and bursting). These omissions can produce detectable model mismatch even when the resulting decoding trajectories remain scientifically informative [5–7].

State space models already contain rich diagnostic information in the form of distributions used to estimate the state process at each time, including the normalized likelihood, which reflects information from only the current neural observations, and the prediction distribution, which aggregates information from all past observations based on the state space model structure. One potential solution for more targeted local diagnostic information would be to directly compare these distributions. We do not expect these two distributions to be identical. The prediction distribution aggregates information from many past spikes; the likelihood reflects only the current observation. The two distributions therefore carry different amounts of information about the state and will typically have different shapes and spreads. What matters for model assessment is not whether the distributions differ, but whether they are consistent, meaning that they concentrate their probability mass over overlapping regions of the state space rather than disjoint ones. When these

distributions provide inconsistent information about the state, the mismatch can reveal where, when, and potentially how the model fails to capture the structure of the data. Examining these discrepancies could therefore provide a direct, interpretable measure of model–data agreement—analogueous to a local, time-resolved goodness-of-fit test.

To address these gaps, we develop and validate a suite of local goodness-of-fit methods, each built on this comparison between the prediction distribution and the likelihood. The most familiar way to compare two distributions is to measure their divergence, and therefore we include KL divergence as a natural point of reference. However, divergence measures quantify how much two distributions differ, not whether they are consistent, so they flag any mismatch, including broad prediction distributions or narrow likelihoods even when the models capture the data structure well. We therefore develop two consistency-focused measures, the highest posterior density (HPD) region overlap and a predictive check measure, and show that they flag genuine model–data inconsistency and provide actionable feedback to neuroscientists. Using simulated and real hippocampal data, we demonstrate how each metric responds to different types of misspecification and argue that consistency is the right target for local goodness-of-fit assessment. This approach offers a path for neuroscientists to move from binary model rejection (is this model good or bad?) toward an iterative, diagnostic approach to building interpretable theories of neural dynamics.

Methods

State-space models

The state-space paradigm is particularly well suited for neural data analysis because many neuroscience experiments focus on internal brain processes that are not directly observable. For example, hippocampal replay of spatial trajectories has been hypothesized to be related to mental exploration of space, but the trajectories being replayed are not directly observable by experimentalists [6]. In such cases, state-space models are used to simultaneously estimate the unobserved internal representations and their relation to neural activity patterns.

For a discrete set of time points, $t_0, \dots, t_K$, we define an unobserved state process x_k , which influences a neural observation process y_k at each time t_k . We define a state space model by combining a state transition distribution $p(x_k | x_{k-1})$, which defines how the state evolves in time, with an observation likelihood $p(y_k | x_k)$, which defines the relationship between the neural activity and the state. To simplify the problem, we assume that the state transition distribution depends only on the most recent past value of the state and that the observation likelihood depends only on the current state value. Fig 1a shows a schematic of this state-space model structure.

This state-space model structure allows for iterative estimation of the state process at each time step based on all the neural data up to that time (Fig 1b). If the distribution of the state at time t_{k-1} has been computed, we can convolve it with the state transition distribution to obtain a one-step prediction distribution, $p(x_k | y_{1:k-1}) = \int_{x_{k-1}} p(x_k | x_{k-1}) \cdot p(x_{k-1} | y_{1:k-1}) dx_{k-1}$. This predictive distribution combines the information from all previous observations to compute the distribution of the state at the next time step. We update this predictive distribution with the information from the most recent observation by multiplying by the observation likelihood to compute the posterior distribution of the state given all of the data up to the current time step: $p(x_k | y_{1:k}) \propto p(y_k | x_k) \cdot p(x_k | y_{1:k-1})$.

Common approaches for assessing the goodness-of-fit of state-space models include examining the cross-validated error measures in the state process when the state is observable [8], cross-validated error measures in the observation process such as the mean-squared prediction error [9], and the distribution of prediction residuals [10]. Although these approaches can be useful, they may be difficult to interpret for neural data because intentionally simplified state-space models often exhibit some detectable lack of fit even when they remain scientifically useful.

Here, we take a different approach that relies on a set of local goodness-of-fit measures. Each measure compares the one-step predictive distribution with either the normalized observation likelihood as a function of state or the observation itself. These measures therefore consider the consistency between information accumulated through the state-space model over past spiking and the information from the most recent spiking activity. Critically, we do not expect these

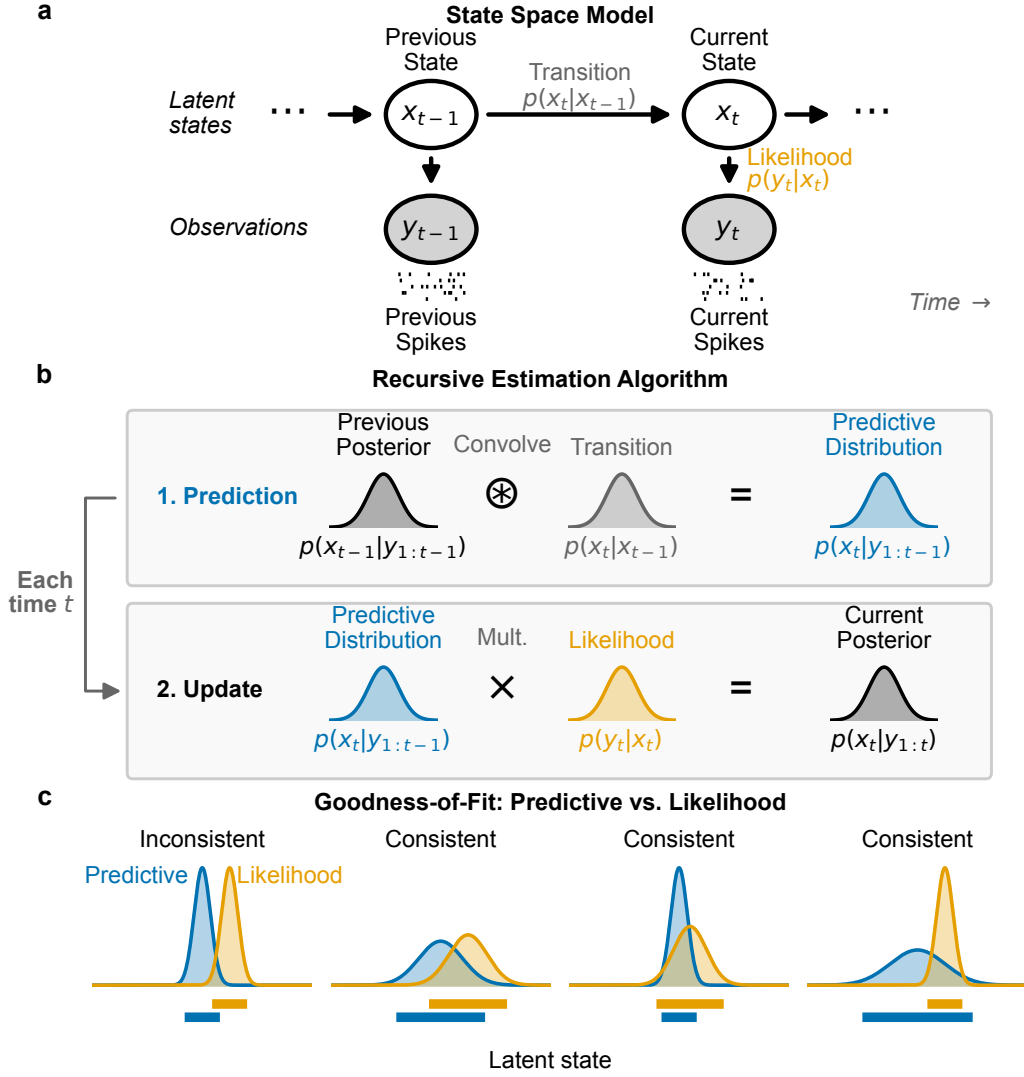


Fig 1. State-space model and local goodness-of-fit measures. **(a)** Graphical model depiction of state-space model includes state transition model, $p(x_k | x_{k-1})$, and observation likelihood $p(y_k | x_k)$. **(b)** These model components are sufficient to compute an iterative expression for the posterior filter distribution at time t_k , $p(x_k | y_{1:k})$ from the distribution at the previous time step. This involves first convolving the previous posterior with the state transition distribution to compute the one-step predictive distribution, $p(x_k | y_{1:k-1})$, and then updating the predictive distribution with information from the most recent spiking by multiplying by the observation likelihood. **(c)** For each new spike observation, we compute the local goodness-of-fit by assessing the consistency of information between the predictive distribution and likelihood. We consider the distributions inconsistent when they assign high probability to mostly disjoint subsets of the state-space. When the regions with high probability overlap, the distributions are considered consistent, even if the distributions differ in their central tendency or certainty.

distributions to align perfectly since they likely contain different amounts of information about the state process. Instead, we use the operational notion of consistency illustrated in Fig 1c. We consider the distributions inconsistent when their probability masses are concentrated in mostly non-overlapping regions of the state space. We consider them compatible when their high-probability regions largely overlap, even if their central tendencies differ. We also regard the distributions as consistent when a broad high-probability region of one distribution contains a substantially narrower region of the other. This situation may arise frequently because the predictive distribution contains information accumulated from many previous spikes.

Local goodness-of-fit measures

In this section, we introduce three classes of metrics designed to characterize the local goodness-of-fit of the state-space model.

Overlap between HPD regions

One way to assess the consistency between the predictive distribution and the observation likelihood is to examine the overlap between their highest posterior density (HPD) regions. We refer to the resulting measure as HPD overlap. A substantial intersection indicates that the distributions assign high density to some of the same state values. Minimal or no overlap indicates disagreement under this criterion. This metric provides an interpretable geometric summary of local consistency between the distributions.

We define the overlap between the HPD for the predictive distribution and the normalized observation likelihood as

$$\text{HPDOverlap}(P, Q) = \frac{|\text{HPD}_P \cap \text{HPD}_Q|}{\min\{|\text{HPD}_P|, |\text{HPD}_Q|\}}, \quad (1)$$

where HPD_P and HPD_Q are the highest posterior density regions of the predictive density and the normalized observation likelihood respectively, and $|\cdot|$ denotes the volume of the region. The

numerator is the size of the overlap between these regions and the denominator is the size of the smaller of the two HPD regions. This normalization ensures that the resulting value is in the range $[0, 1]$, with 0 indicating no overlap, and 1 indicating that one region is entirely contained within the other. For example, if the likelihood region is wider and therefore less informative than the predictive region but completely contains it then the HPD ratio will be 1, suggesting that the two distributions are statistically consistent. Conversely, if the ratio is close to 0, it indicates little or no overlap, suggesting significant differences between the range of values for the state that would be inferred from these two distributions. Throughout, we compute 95% HPD regions.

Fig 2a illustrates the computation of the HPD overlap. We compute high-density regions for the predictive distribution (top) and the normalized observation likelihood (middle), then divide the volume of their intersection by the volume of the smaller region (bottom).

Predictive checks

The HPD overlap compares two distributions of the state: one based on the accumulated past information and the other based on the isolated information from the likelihood of the most recent spiking observation. Another approach is predictive checking, which compares the observation itself to a predictive distribution of that observation estimated from the model. This approach has been used before in other contexts ([11–13]) and has the advantage that it allows for the computation of p -values for individual observations, which are easy to interpret [14, 15].

Predictive checks for spiking data can be challenging to implement, since within sufficiently short intervals, a spike is typically less probable than no spike. One possible approach is to consider spike patterns over longer fixed intervals in which multiple spikes are expected. Here, we take a simpler approach of focusing only on spike times and measuring the likelihood that an observed spike comes from a particular neuron (for spike-sorted models) or has a particular set of waveform features (for clusterless models).

In particular, let $S_{k,j} = 1$ indicate that the j th spike in the k th time bin occurred, and let $y_{k,j}$ denote its mark: the identity of the neuron for a spike-sorted model or a waveform feature for a

clusterless model. Because the analysis is conditioned on a spike event, the relevant observation distribution is the distribution of its mark given that a spike occurred. For a spike-sorted model with conditional intensity $\lambda_c(x_k)$ for neuron c , this distribution is

$$q_c(x_k) = p(y_{k,j} = c \mid x_k, S_{k,j} = 1) = \frac{\lambda_c(x_k)}{\sum_{d=1}^C \lambda_d(x_k)}. \quad (2)$$

The corresponding clusterless quantity is the conditional density of the waveform mark given a spike. We define the predictive distribution of the mark based on the state model as

$$f_{\text{pred}}(y_{k,j}) = \int p(y_{k,j} \mid x_k, S_{k,j} = 1) p(x_k \mid y_{1:k-1}) dx_k, \quad (3)$$

where $p(y_{k,j} \mid x_k, S_{k,j} = 1)$ is the conditional mark distribution (Eq. (2) for spike-sorted data) and $p(x_k \mid y_{1:k-1})$ is the one-step prediction distribution of the current state given all previous spikes. We can sample new observations $\tilde{y}_{k,j}$ from this predictive distribution, and compute the probability that the observed spike is at least as likely as the sampled spike:

$$p_{k,j} = \Pr_{\tilde{y}_{k,j} \sim f_{\text{pred}}} (\log f_{\text{pred}}(y_{k,j}) \geq \log f_{\text{pred}}(\tilde{y}_{k,j})). \quad (4)$$

The log terms in Eq (4) are unnecessary, but suggest the method to estimate $p_{k,j}$. We sample x_k multiple times from the one-step prediction distribution, $p(x_k \mid y_{1:k-1})$, and use these to generate samples of the spike observations, $\tilde{y}_{k,j}$, from the predictive distribution. We then compute the fraction of these samples whose log likelihood is smaller than or equal to that of the observed spike. For the spike-sorted models considered here, $\tilde{y}_{k,j}$ takes one of a finite set of values—the identities of the recorded neurons—so $p_{k,j}$ can be computed exactly by summing f_{pred} over all cell outcomes at least as unlikely as the observed spike, rather than by sampling. Because $p_{k,j}$ ranks the observed spike against this predictive distribution of possible spikes, we refer to it as the *rank-based predictive p-value*.

Fig 2b illustrates the computation of a predictive p -value for spike-sorted data. Many values

of the state are sampled from the predictive distribution (top), each of which is used to sample a neuron that could have generated the observed spike (middle). The log likelihood of the observed spike is then compared to an empirical distribution of the log likelihoods of the sampled spikes (bottom) leading to a goodness-of-fit metric that can be interpreted as a p -value for that individual spike. High p -values suggest that the observed spike is consistent with the predictive distribution of spikes and a p -value near zero suggests that the observed spike is unexpected based on the predictive distribution computed from the past spiking and the model.

While the HPD overlap ratio was explicitly designed to satisfy our definition of consistency between the predictive distribution and the rescaled likelihood in the state space, the predictive check inherits the same consistency property, even though it operates on the observations directly, through the marginalization in Eq. (3). We expect to see large p -values when these distributions match, but also when either distribution is narrower than the other as long as they point to overlapping regions of the state space. In the common case that the predictive distribution of the state is narrow relative to the likelihood, the predictive distribution of observations will be concentrated on the spikes that occur at the confidently predicted location, and the observed spike will have appreciable likelihood at that location. The observed spike is among those that the model anticipates and should therefore lead to a high p -value. In the less common case that the likelihood is narrow relative to the predictive distribution of the state, the predictive distribution of the observations will also be broad and will assign appreciable probability to a wide range of spikes. The observed spike is likely to fall within this anticipated range, so the p -value should also be high. A low p -value should arise only when the observed spike is specifically unlikely relative to what the model anticipates, which corresponds to the likelihood concentrating in a region the accumulated past deems improbable.

KL divergence

Finally, one of the most well-known measures of discrepancy between probability distributions is the Kullback–Leibler (KL) divergence [16], defined in this case as the expected excess logarithmic loss incurred when the normalized likelihood is used to approximate the predictive distribution, with

the expectation taken with respect to the predictive distribution:

$$D_{\text{KL}}(P \parallel Q) = \int P(x_k) \log\left(\frac{P(x_k)}{Q(x_k)}\right) dx_k, \quad (5)$$

where $P(x_k) = p(x_k \mid y_{1:k-1})$ is the predictive distribution and $Q(x_k) = p(y_k \mid x_k) / \int p(y_k \mid x_k) dx_k$ is the normalized likelihood in x_k given the observed spiking y_k .

Fig 2c illustrates the computation of the KL divergence. The top panel shows the predictive distribution and normalized observation likelihood as functions of x_k . The middle panel shows their pointwise log-density ratio, $\log(P(x_k)/Q(x_k))$. The bottom panel shows this ratio weighted by $P(x_k)$, which, when integrated, gives the value of the KL divergence.

The KL divergence is the most familiar way to compare two distributions, and we include it as a natural point of reference. It is an asymmetric measure, meaning that $D_{\text{KL}}(P \parallel Q) \neq D_{\text{KL}}(Q \parallel P)$. We could also compute the KL divergence from the likelihood to the predictive distribution, $D_{\text{KL}}(Q \parallel P)$, or the average of these two divergences, $\frac{1}{2}(D_{\text{KL}}(P \parallel Q) + D_{\text{KL}}(Q \parallel P))$, but both alternatives lead to a common issue, namely that the KL divergence can become arbitrarily large when its first argument places appreciable probability where its second argument assigns very little. Thus, $D_{\text{KL}}(Q \parallel P)$ can become large when an uninformative spike produces a broad likelihood. Choosing $D_{\text{KL}}(P \parallel Q)$ avoids that particular behavior but raises a complementary one: the divergence will grow large whenever the predictive distribution is broad relative to the likelihood, which can occur, for example, whenever there are sparsely firing, informative spikes. We will see in the simulation and data analysis below that this flags many spikes, even when their information is compatible with the accumulated information from past spiking.

Simulation study

Validating local goodness-of-fit metrics on real neural data is fundamentally limited: when the truth is unobserved, a metric that flags a moment of model failure cannot be distinguished from one that flags a moment of unusual neural activity. We therefore first evaluate the three diagnostics—HPD

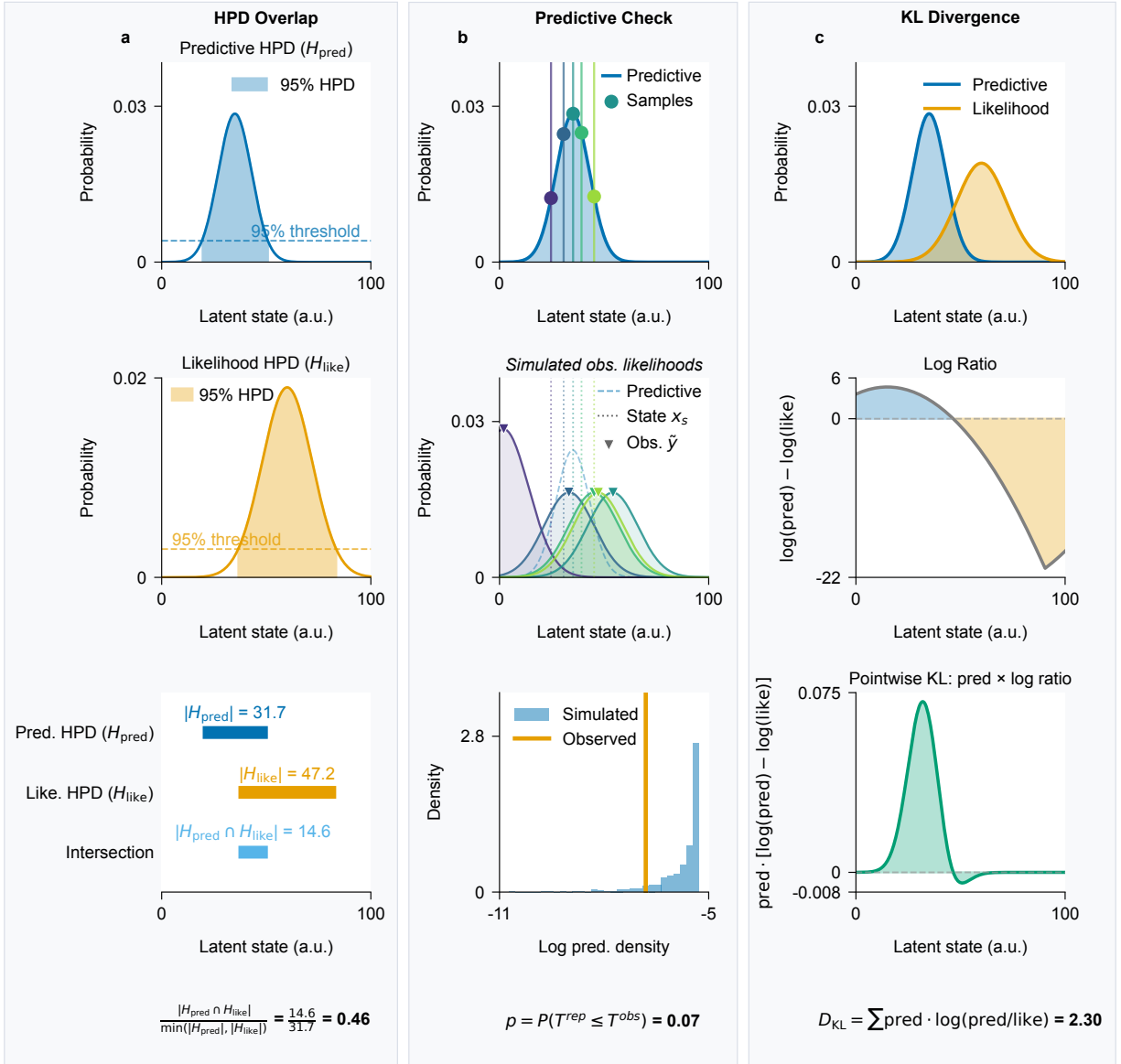


Fig 2. Computation of the three local goodness-of-fit metrics. **(a)** For HPD overlap, high-density regions are computed for the predictive distribution (top) and the normalized observation likelihood (middle). The volume of their intersection is then divided by the volume of the smaller region (bottom). **(b)** For the predictive check, state values are sampled from the predictive distribution (top), and each sampled state is used to generate the identity of a neuron that could have produced the spike (middle). The metric is the probability that the log predictive probability of the observed spike is greater than or equal to that of a replicated spike (bottom). **(c)** The KL divergence compares the predictive distribution (blue) and the observation likelihood (orange) (top) by computing the difference in their log density (middle) and computing its expectation with respect to the predictive distribution (bottom).

overlap, the rank-based predictive p -value, and KL divergence—in a controlled benchmark in which each perturbation is known by construction. The benchmark simulates a hippocampal place-cell population on a one-dimensional track and runs a Bayesian state-space decoder over the full session. Three windows deliberately perturb either the generative process or the decoder’s assumptions; two additional controls test unusual but internally consistent neural activity. Together, these conditions distinguish model misfit that leads to decoding errors from differences in distribution shape that do not imply conflicting state information. This relates to our original goal of identifying local periods where the information from the current spiking is inconsistent with the aggregated information from our model based on past spiking.

Generative model

The simulation applies a Bayesian filter to a 32-second session containing three misspecification windows, three clean-recovery windows, and two internally consistent controls.

Track and observation model. The state space is a one-dimensional linear track of length $L = 100$ arbitrary units (a.u.) discretized at unit spacing, $x \in \{0, 1, \dots, 100\}$. The simulated population consists of eleven place cells with Gaussian place fields centered at $\mu_c \in \{0, 10, \dots, 100\}$ and shared standard deviation $\sigma_{\text{pf}} = 10$ a.u. The expected spike rate of cell c at position x is

$$\lambda_c(x) = \alpha \phi(x \mid \mu_c, \sigma_{\text{pf}}^2), \quad (6)$$

where $\phi(\cdot \mid \mu, \sigma^2)$ is the Gaussian density and $\alpha = 5$ is a peak-rate scale factor. With a 1 ms simulation step, the peak rate is $\alpha/(\sigma_{\text{pf}}\sqrt{2\pi}) \approx 200$ Hz. Spike counts are drawn independently across cells at each step, $y_{c,t} \sim \text{Poisson}(\lambda_c(x_t))$.

Five additional cells form a sparse population of narrow fields clustered near $x = 30$ a.u. (centers spread over 28.5–31.5 a.u., each with standard deviation 2 a.u.). Each fires as an independent Poisson process. Its per-cell peak rate is 0.01 Hz during the baseline and recovery periods and 1 Hz

during the final sparse-population control; with five cells the aggregate rate stays low (a few spikes per second), so the spikes remain temporally isolated. The decoder uses these same epoch-specific rates, so recruiting these cells does not itself create an observation-model mismatch.

Position dynamics. During baseline windows, the animal’s position evolves as a Gaussian random walk with step standard deviation $\sigma_{\text{pred}} = 0.5$ a.u. The trajectory is reflected at 0 and 100 a.u. throughout the session, starts at $x_0 = 0$, and is continuous across phases: the final position of each phase becomes the initial position of the next.

Phase schedule. The 32-second simulation ($T = 32,000$ steps) is partitioned into eight contiguous phases: an opening baseline, three misfit windows, three clean-recovery interludes, and a final sparse-population control. Each misfit window perturbs the observation model $p(y_k | x_k)$ or the state-transition model $p(x_k | x_{k-1})$ by altering either the generative process or the decoder’s assumptions:

Remap: During steps 6,000–10,000, the decoder-side observation model is perturbed. The decoder evaluates per-spike likelihoods against a globally remapped set of place-fields: the eleven place-field centers are scrambled by a fixed random permutation (a derangement in which every field is displaced by at least three positions), modeling global remapping to statistically independent locations. Both the decoder’s posterior update and its per-spike diagnostics use this remapped likelihood—the diagnostics are evaluated against the model the decoder actually uses, not the true generative fields—so the misfit surfaces from the scramble’s spatial incoherence rather than from any oracle knowledge of the correct fields. Spike generation is unchanged.

History-dependent firing: During steps 14,000–18,000, the generative-side observation model is perturbed. Spiking is generated with a 1 ms hard refractory period followed by a 2–10 ms burst window with a threefold rate increase, matching the rough phenomenology of CA1 pyramidal-cell refractory and burst windows; the decoder still assumes independent Poisson

firing. The per-spike spatial likelihood is unchanged, so the misfit lives in the spike-train temporal joint distribution.

Drift: During steps 22,000–26,000, the generative-side state-transition model is perturbed. The trajectory is generated with persistent velocity, $v_t = 0.88 v_{t-1} + \eta_t$ and $x_t = x_{t-1} + v_t$ with $\eta_t \sim \mathcal{N}(0, \sigma_{\text{pred}}^2)$; the decoder continues to assume a memoryless random walk with step standard deviation σ_{pred} .

Sparse-population control. During steps 30,000–32,000, the animal is immobile at $x = 30$ a.u. and the ordinary place-cell ensemble becomes quiet, as can occur during immobility. The clustered sparse-population cells increase to their 1 Hz peak rate and fire intermittently. Unlike the misfit windows, this is an internally consistent control: the decoder knows both the quiet ensemble rate and the active sparse-population rates, and decodes with its usual random-walk motion prior. Because that prior expects the animal to move, in the intervals between isolated spikes the predictive distribution gradually spreads; each sparse-population spike then supplies a narrow likelihood centered inside that prediction. The two distributions therefore remain consistent by the HPD-overlap and predictive-check criteria, while their large difference in concentration can elevate KL divergence.

The three intermediate clean-recovery windows (steps 10,000–14,000; 18,000–22,000; 26,000–30,000) restore both the generative process and the decoder to the baseline configuration, providing out-of-sample well-specified comparison periods. They are summarized in Fig. 3b but are not used to set the diagnostic thresholds, which are derived from the opening baseline. During the last second of the final recovery, the animal moves gradually to the fixed location so the sparse-population control begins without an abrupt position jump. With this schedule fixed, each diagnostic can be evaluated against the known model faults and the two internally consistent controls.

Replay control. Embedded in the second clean-recovery window (steps 19,000–21,000) is a replay event: the animal is immobile while a coherent trajectory sweeps out and back across the

track. The decoder’s observation and motion models are consistent with the replayed spikes, so it faithfully tracks the swept trajectory—its decoded position departs markedly from the animal’s true (fixed) position while the predictive distribution and each spike’s likelihood remain in agreement throughout the sweep. Because the diagnostics assess the internal consistency of the decoded state rather than its correspondence to the animal’s physical position, this large decoded-versus-true divergence is correctly left unflagged: it reflects a change in what the ensemble represents, not a failure of the model. The replay steps are excluded from the pooled well-specified comparison column and are scored as their own column in Fig 3b.

Simulation results

Fig 3a shows a realization of the simulation, the decoding results, and the three local goodness-of-fit metrics. The top panel shows the true position of the simulated rat in magenta and the predictive decoding distribution in black. The regions with white backgrounds are the clean-recovery periods; colored backgrounds indicate the three model-misfit periods and two controls. The second panel shows the normalized observation likelihood as a function of position at each spike time. The third panel shows the spiking from each of the sixteen simulated neurons. Since these are sorted by their place field centers, the pattern of spiking aligns with the likelihood as a function of position. Note that the remapped period is the one in which the predictive distribution and the spike likelihoods most clearly disagree about the animal’s location. During the replay control, by contrast, the predictive distribution departs strongly from the true position (the decoder tracks the replayed trajectory) yet continues to agree with the spike likelihoods, so the divergence is benign. The sparse-population control is visibly different: after the ordinary ensemble falls silent, isolated spikes appear only in the clustered sparse-population cells.

The lower three panels of Fig 3a show, from top to bottom, the values of the HPD overlap, the rank-based predictive p -value (as $-\log(p)$) and KL divergence for each spike. In each panel a horizontal line marks the threshold used to flag poor fit. Because low HPD overlap indicates disagreement, a spike is flagged when its value is at or below the 1st percentile of values from the

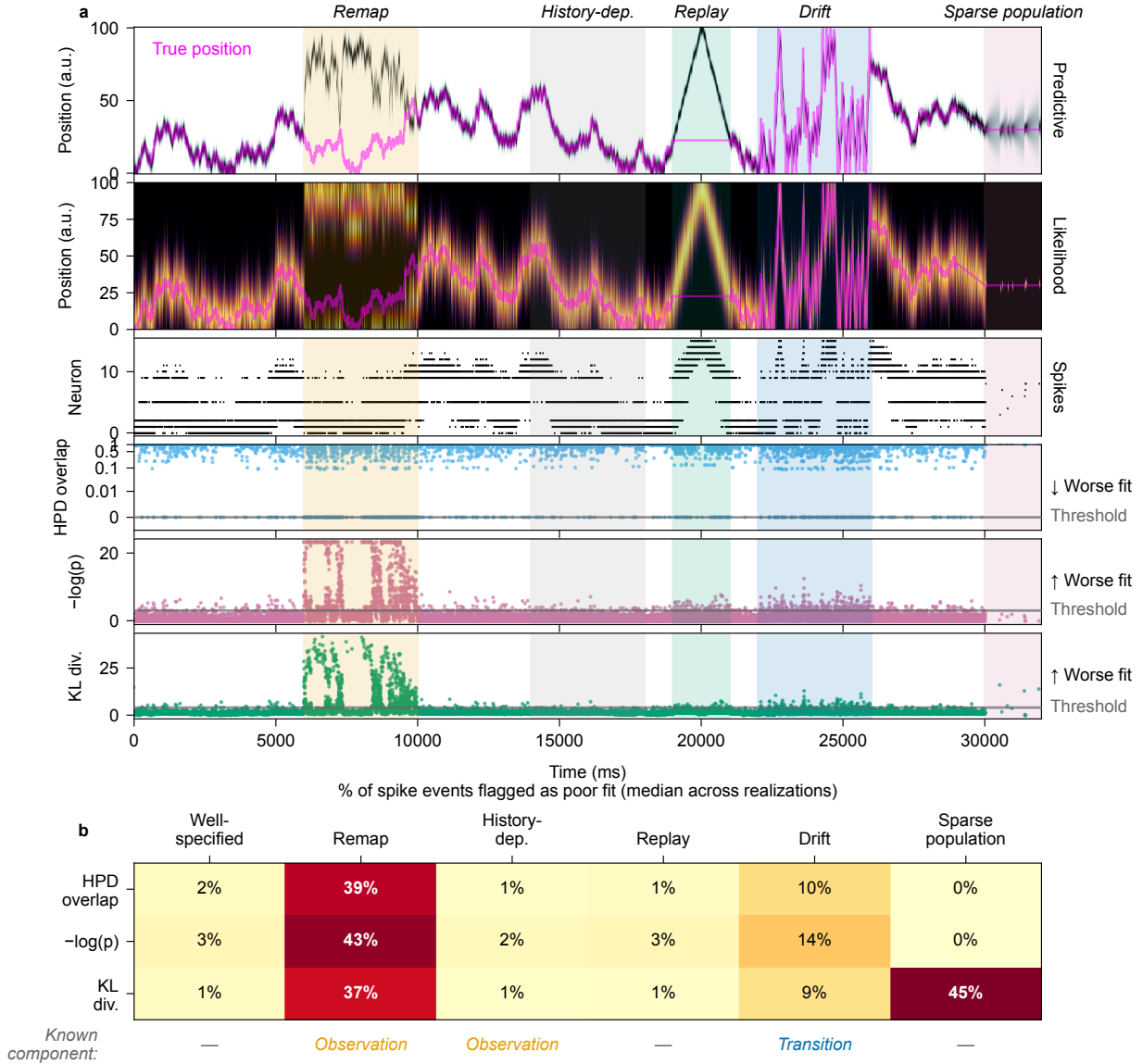


Fig 3. Per-spike goodness-of-fit diagnostics in a simulation with three types of model misspecification separated by clean-recovery windows—place-field remapping, unmodeled history-dependent firing, and persistent-velocity drift—together with replay and sparse-population controls. **(a)** The top row shows the predictive distribution as a heatmap of time and position. The position is overlaid as a magenta line. The three misfit periods and two controls are shaded in color. The second row from the top shows the per-spike likelihood as a heatmap, restricted to time bins containing at least one spike, with position again overlaid in magenta. The third row shows the spike raster with neurons sorted by place-field peak position. The bottom three rows show HPD overlap, the rank-based predictive p -value, and KL divergence for each spike event, with a horizontal line marking the threshold used to flag poor model fit; the HPD overlap is plotted on a symmetric-log y -axis to resolve values near zero. **(b)** The percentage of spike events flagged by each diagnostic in each phase, summarized by the median across 100 independent simulation realizations. The well-specified clean-recovery windows are pooled into a single column for comparison, and each control is scored as its own column. Shared HPD and KL thresholds are derived by pooling per-spike values from the opening baseline (steps 0–6,000) across all 100 realizations; the predictive p -value uses the fixed cutoff of 0.05. Pooling stabilizes the thresholds and per-phase percentages, which are noisy when estimated from a single run. A row beneath the heatmap identifies the model component perturbed in each phase—the observation model (*Observation*) or the state-transition model (*Transition*)—and uses a dash for the well-specified and control phases.

opening baseline (steps 0–6,000) pooled across 100 independent simulation realizations. For the rank-based predictive p -value, a spike is flagged when $p \leq 0.05$ (equivalently, $-\log(p) \gtrsim 3$), a fixed cutoff rather than a baseline-derived threshold. Because high KL divergence indicates disagreement, a spike is flagged when its value is at or above the 99th percentile of values from the same pooled opening-baseline distribution. Pooling across realizations stabilizes the baseline-derived HPD and KL thresholds, which would otherwise be noisy when estimated from a single run.

We see that all three metrics agree that there is a lack of consistency during the remapping period, where the likelihood from the remapped cells is in a different region of the environment than the model estimate based on the incorrect place field estimates. During the period when the history dependence of spiking is misspecified, the estimated decoding distribution becomes narrower but remains consistent with the spike likelihoods, and none of the three metrics flag this period as leading to incorrect estimates. The replay control is likewise unflagged, though for a different reason: although the decoded position sweeps far from the animal’s true location, the predictive distribution tracks the replayed trajectory and remains consistent with each spike’s likelihood, so all three metrics stay at their baseline levels. During the period when the decoder misses a drift term, the actual movement trajectory has notably increased velocity. The decoder only partially keeps up, so its predictive estimates lag behind the true location; this lag biases the predictive spiking distribution toward neurons consistent with the lagged position. As a result, the diagnostics flag a modest percentage of spikes during the drift period—most clearly the rank-based predictive p -value—though far less than during the remapping period. During the sparse-population epoch, the ordinary ensemble is quiet and the prediction broadens naturally between intermittent spikes from the clustered cells. Each such spike concentrates inside the range of positions already supported by the prediction. HPD overlap and the rank-based predictive p -value therefore classify the two distributions as consistent, whereas KL divergence responds to their large difference in concentration. This familiar sparse-firing regime illustrates why distributional difference need not imply conflicting state information.

Fig 3b shows the percentage of spikes flagged by each metric in each period, summarized by

the median across 100 independent realizations of the simulation. A spike is flagged whenever its diagnostic crosses the corresponding threshold in the direction of worse fit: at or below the 1st-percentile baseline value for HPD overlap, at or below the fixed 0.05 cutoff for the rank-based predictive p -value, or at or above the 99th-percentile baseline value for KL divergence. The median percentage flagged during the remapping period is substantial for all three metrics (37–43%), demonstrating that all three methods are able to identify inconsistent spiking information leading to incorrect state estimates; the percentage nonetheless varies from realization to realization because it depends on how much of the scrambled map the simulated trajectory visits. All three metrics flag only 1–2% of spikes during the history-dependence period. The replay control is flagged at a low rate—a median of 1–3% across the three metrics, comparable to the well-specified baseline—confirming that a large decoded-versus-true divergence is not, by itself, registered as model misspecification. The drift period is flagged more often (9–14%), most notably by the rank-based predictive p -value, but still far less than the remapping period. In the sparse-population control, HPD overlap and the rank-based predictive p -value flag no spikes, whereas KL divergence flags a median of 45%. During the quiescent intervals between isolated sparse-population spikes the decoder’s estimate receives little new information and its prediction broadens; each spike then arrives as a sharply localized, spatially informative observation whose location remains well inside that broad prediction. HPD overlap and the predictive check therefore register the spike and the prediction as consistent, while KL responds to their large difference in concentration. This familiar sparse-firing regime—the kind that arises during immobility—illustrates that a distributional difference between prediction and likelihood need not reflect conflicting information about the state.

Real hippocampal data analysis

Data collection and behavioral task

We applied the local goodness-of-fit metrics to units recorded from the rat hippocampus while the animal performed a spatial bandit task on a maze with three radially arranged Y-shaped foraging

patches extending from a central point. A schematic of the maze is shown in the right inset in Fig 4. The details of this task and the data collection methods are described by Comrie et al. [17]. Briefly, the rat was trained to run between six reward ports located in three Y-shaped "patches", each with a different probability of reward. Reward probabilities changed periodically after a preset time or trial limit. Recordings were made with 22 tetrodes advanced to CA1; two additional tetrodes, one in each hemisphere, served as local references in the corpus callosum.

For the present analysis, we sorted the continuous 30 kHz neural signal into putative single units with MountainSort4 [18]. In the MountainSort4 pipeline, the signal was band-pass filtered during preprocessing and whitened; negative-going events were then detected on each tetrode at a threshold of 3 in whitened units, with a minimum inter-event interval of 10 samples (≈ 0.33 ms). Units were retained from automated curation only, without manual cluster curation. Because curation was automated, some units may contain multiunit activity or represent incompletely merged clusters; this does not affect our purpose here, which is to demonstrate the diagnostics rather than to characterize individual neurons. For decoding, we linearized the animal's position by projecting it to the nearest point of a skeletonized track, discretized the resulting linear position into spatial bins of approximately 2 cm, and binned spikes at 2 ms resolution. The Continuous–Fragmented decoder represented the joint state of discrete dynamics mode and position, with the same position-dependent observation model in both modes. For all three diagnostics, we marginalized its one-step predictive distribution over dynamics mode and compared the resulting distribution over position with one copy of the shared observation likelihood. The diagnostics for both decoders were therefore defined on the same position grid. Because the recording has no designated clean baseline period, we used fixed cutoffs—HPD overlap below 0.05 and a rank-based predictive p -value below 0.05—rather than the within-session baseline percentiles used in the simulation.

Data analysis results

Fig 4 compares decoding of these hippocampal spike data under two state-space decoding models. The first assumes that the latent state moves continuously through space, as expected if it represents

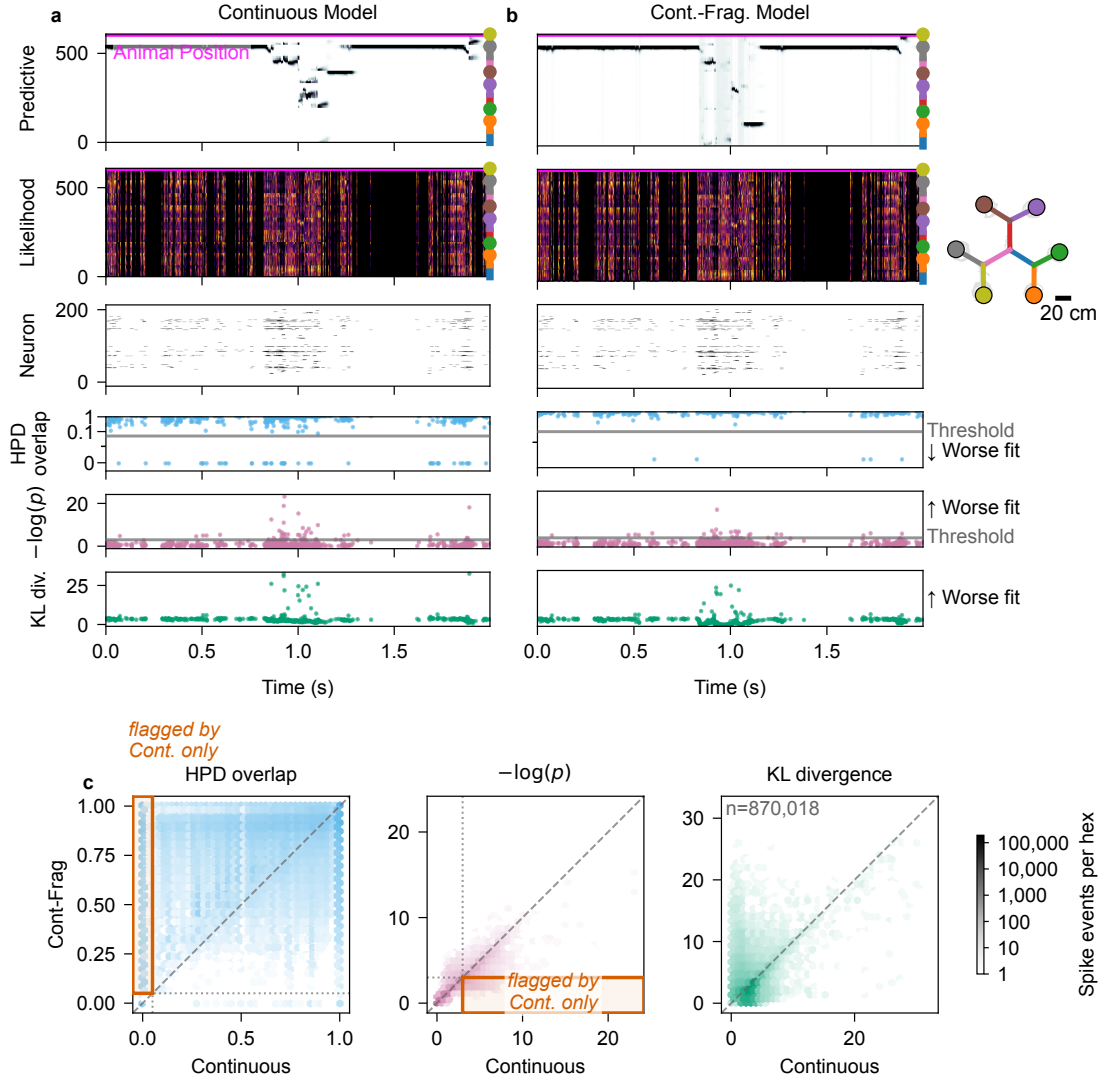


Fig 4. Per-spike goodness-of-fit diagnostics applied to a hippocampal recording containing 203 simultaneously recorded units during navigation on a multi-arm maze, using data from Comrie et al. [17]. Two state-space models—a Continuous model and a Continuous–Fragmented model—are evaluated on the same session. The unlettered track inset shows the two-dimensional maze layout, with arms drawn as colored segments, colored dots marking the six reward wells, the animal’s trajectory overlaid in gray, and a 20 cm scale bar. **(a, b)** A detail window containing a candidate replay event, decoded with the Continuous and Continuous–Fragmented models, respectively. All rows share a common horizontal axis showing time in seconds relative to the start of the window. **Row 1, Predictive:** the one-step predictive distribution over linearized position, with the animal’s physical position overlaid. The discrete dynamics mode of the Continuous–Fragmented model is marginalized so that both models are evaluated over the same position grid. **Row 2, Likelihood:** the combined observation likelihood in each spike-containing time bin. **Row 3, Unit:** a spike raster with units sorted by place-field peak position. **Rows 4–6:** one marker per spike event for HPD overlap, $-\log(p)$, and KL divergence, respectively. For HPD overlap, a horizontal line marks the fixed cutoff of 0.05. For the rank-based predictive p -value, the line at $-\log(0.05) \approx 3$ corresponds to $p = 0.05$. KL divergence has no fixed cutoff and is shown without a threshold. The direction of poorer local fit is labeled at the right. **(c)** Pairwise comparisons of per-spike diagnostic values across the entire session, with the Continuous model on the x -axis and the Continuous–Fragmented model on the y -axis. The dashed line denotes identity. In the HPD overlap and $-\log(p)$ panels, dotted lines mark the corresponding flag thresholds, and the outlined box identifies spikes flagged by the Continuous model but not the Continuous–Fragmented model. Color intensity indicates the number of spike events on a shared logarithmic scale.

only the animal’s physical location. During immobility, however, hippocampal representations can jump to distant locations and reactivate in sequences resembling past movement trajectories [19, 20]. We therefore expect the Continuous model to exhibit local disagreement with the spiking data when the neural representation jumps. We compare this model with a Continuous–Fragmented model that permits discrete jumps in the hippocampal representation. We previously used a related model structure to capture replay and other nonlocal hippocampal representations [2].

Fig 4a shows a brief segment decoded with the Continuous model while the animal is immobile. The panels follow the organization of Fig 3a. During this candidate replay event, the decoded representation transitions rapidly among maze locations while the animal remains at rest, as indicated by the magenta position trace. The Continuous model cannot track these rapid transitions, and its predictive distribution lags behind the information provided by the likelihoods of the observed spikes. The HPD overlap and rank-based predictive p -value diagnostics flag many spikes, and several spikes also have elevated KL divergence; KL divergence values are not dichotomized in this analysis.

The contrast between the Continuous and Continuous–Fragmented models is shown in panels a and b of Fig 4. The likelihood of each spike is identical under the two models (second row), but the Continuous–Fragmented model decodes a rapidly moving state trajectory that the Continuous model cannot track. When the representation is assigned to the fragmented state, the predictive distribution becomes uniform or nearly uniform, allowing the likelihoods of subsequent spikes to produce a jump in the state estimate.

For HPD overlap (fourth row, blue), nearly all spikes flagged under the Continuous model are no longer flagged under the Continuous–Fragmented model. Some spikes also have lower $-\log(p)$ values under the Continuous–Fragmented model than under the Continuous model (fifth row, pink). Under the Continuous model, the predictive distribution can concentrate at the wrong location, making the observed spike very unlikely. During replay events, the Continuous–Fragmented model instead produces a broad predictive distribution. Although the observed spike can remain unlikely under this broad distribution, it is less unexpected than under the concentrated Continuous-model prediction. For KL divergence (bottom row, green), many spikes retain high values under the

Continuous–Fragmented model. This pattern is consistent with the simulation results: predictive distributions that are broad relative to spike likelihoods produce larger KL values even when our operational criterion classifies those prediction–likelihood pairs as consistent.

Fig 4c shows hexagonally binned density plots comparing the per-spike values of each local goodness-of-fit metric under the Continuous and Continuous–Fragmented models; color indicates the number of spike events in each hexagon. The left panel shows HPD overlap in blue. Hexagons above the 45° line correspond to spikes with greater HPD overlap under the Continuous–Fragmented model. The dense vertical band at $x = 0$ and $y > 0$ (within the outlined box) represents spikes for which the Continuous model’s predictive distribution has no overlap with the spike likelihood, whereas the Continuous–Fragmented model’s predictive distribution has nonzero overlap. There is comparatively little density at $y = 0$ and $x > 0$, indicating that adding the fragmented state alleviates this disagreement for many spikes. Of the 18,328 spikes flagged under the Continuous model, 16,873 (92%) are no longer flagged under the Continuous–Fragmented model (*rescued*), whereas 173 spikes are flagged only under the Continuous–Fragmented model (*newly flagged*).

The center panel of Fig 4c shows the negative logarithm of the rank-based predictive p -value, $-\log(p)$, in pink. Hexagons below the 45° line correspond to spikes that are less unexpected under the Continuous–Fragmented model than under the Continuous model. The density band below this line extends beyond $x = 3$ (within the outlined box), corresponding to $p < 0.05$ because $-\log(0.05) \approx 3$. The corresponding region above the line, where the Continuous–Fragmented value exceeds 3, is substantially less dense. Thus, the rank-based predictive p -value identifies many spikes for which local disagreement is reduced under the Continuous–Fragmented model. Of the 33,627 spikes flagged under the Continuous model, 8,626 (25.7%) are rescued by the Continuous–Fragmented model, whereas 1,756 spikes are newly flagged.

The right panel of Fig 4c shows KL divergence values in green. Hexagons below the 45° line correspond to spikes with lower KL divergence under the Continuous–Fragmented model. A dense concentration of spike events lies below the line for Continuous-model values below 5. More strikingly, a region above the line contains Continuous-model values near 2 and Continuous–

Fragmented values extending to approximately 20. As in the simulation, these values occur when the Continuous–Fragmented model produces a broad predictive distribution after estimating a fragmented state. KL divergence therefore aligns less closely with the nesting-tolerant consistency criterion used for the other metrics in this setting.

Discussion

State space modeling has become a powerful tool for describing the latent dynamics of neural systems. Validating these models is inherently difficult because latent states are unobservable and many models are purposefully simplified. To address this, we developed multiple local goodness-of-fit metrics that compare the instantaneous information received from each spike to the aggregated information computed based on recent spiking activity and the state-space model assumptions. Rather than dismissing a model in its entirety, these metrics are able to assess the consistency between immediate observations and the information accumulated through the state-space structure, providing a more granular assessment of decoding reliability.

Two of the three metrics we examined, the HPD overlap ratio and predictive checks, met our criteria for assessing the consistency between the information in the predictive distribution and the likelihood. The HPD overlap offers a robust geometric measure of similarity, consistently maintaining lower error rates across various conditions. Predictive checks provide a direct statistical assessment through observation-level p -values, but are more computationally intensive. The KL divergence, in contrast, is computationally simple but is sensitive to distribution tails and lacks a clear upper bound, making it difficult to assess similarity. More broadly, KL divergence is only one member of the larger family of f -divergences, which also includes the total variation distance [21] and the Pearson χ^2 divergence [22]. Substituting another member of this family would emphasize different features of the prediction-likelihood mismatch, but because these measures all target distributional difference rather than the consistency we seek, KL’s tendency to flag spikes that remain consistent with the model, as occurs whenever the predictive distribution is broad, would

likely carry over to the family as a whole. The KL divergence (or any of these generalizations) may be useful in cases when the goal is to flag time points where the predictive distribution becomes uncertain relative to the information coming from individual spikes. When the goal is consistency in estimates of the state-process, we recommend the HPD overlap and predictive checks as primary diagnostic measures.

We illustrated the application of these metrics to simulated and real spike-sorted data from rat hippocampus. These same metrics naturally extend to other neural data types when incorporated into state space models. In particular, recent work has shown that clusterless neural spiking models can substantially reduce bias and variance in neural decoding and estimation problems when compared to spike-sorted models [23]. The metrics, as defined in the [Methods](#) section, naturally extend to clusterless models. In that case, a lack of fit for any particular spike suggests that a spike with that waveform was unexpected based on the accumulated information from the state space model up to that point in time.

These metrics make it possible to identify specific periods of time where a state-space model is more or less consistent with the spiking data. For example, we show that a model that includes a fragmented state that allows jumps in hippocampal representations provides a better fit to many spikes than a continuous-only model where these jumps are highly unlikely. More broadly, these approaches enable a feedback loop of model development and model assessment that could greatly improve the quality of our state-space models.

There are a number of ways we could extend these methods. For simplicity, we focused on comparing instantaneous information in the likelihood of each spike with aggregated information from the past contained in the one-step prediction density. A natural and powerful extension of these methods is obtained by replacing the one-step prediction density with any posterior density derived from the state-space framework. In particular, replacing the one-step prediction density with a smoothing density (e.g., from a fixed-interval smoother over the observation period) would allow us to leverage both past and future data to increase statistical power in identifying model misspecification.

This potential impact of these approaches does come with challenges, however. First, the computational burden of computing these metrics increases as the dimensionality of the state process increases. In all of our examples, we computed the KL divergence measure and the HPD overlap by partitioning the state space into a lattice and computing the likelihoods and prediction densities at each point in that lattice. As the dimension of the state increases, the size of that lattice will grow exponentially. While the predictive checks are the most computationally burdensome in lower state dimensions, the fact that we are sampling from the state-space rather than computing values over a lattice means that in higher dimensions the predictive checks will become more computationally efficient. It also suggests that we may be able to improve the efficiency of the KL and HPD overlap metrics by using sampling-based estimation methods. This will not solve the curse of dimensionality, as the number of samples required for accurate estimates will likely still grow with dimension, but it is likely to allow for efficient estimation in moderate dimensions. Another approach for dealing with this issue may be to compute these consistency metrics in lower-dimensional subspaces of the state space. Identifying subspaces that retain the statistical power of the original space is a topic for future exploration.

Another potential challenge relates to identifying time-scales for which to assess for lack of fit. These methods allow us to focus on time-scales as short as individual spikes, and for our place cell decoding examples individual spikes are sufficiently informative that we are often able to detect local model misfit at this time-scale. For other neural systems, it may be the case that local model errors are more likely to be detected over longer time-scales. In this paper, we noted that these goodness-of-fit measures can be extended to longer time intervals, but determining what time-scales are most appropriate may require relying on prior knowledge about the system, or exploring the values of these metrics across multiple time-scales.

We believe that these local goodness-of-fit tools provide an important step toward making inferences from state-space neural models more reliable and interpretable. By pinpointing specific moments when the model assumptions break down, researchers can be more confident in their inferences. Ultimately, these measures empower us to refine our understanding of neural dynamics

and deepen our understanding of how the brain encodes and processes information.

Data availability

The hippocampal recording analyzed here was originally reported by Comrie et al. [17] and is available on the Distributed Archives for Neurophysiology Data Integration (DANDI) Archive as dandiset 001942 (<https://dandiarchive.org/dandiset/001942>).

Code availability

All simulation, analysis, and figure-generation code is publicly available at <https://github.com/edeno/statepacecheck-paper>. [TODO: Add the archival DOI (e.g., Zenodo) for the tagged release used in this paper.]

Ethical statement

All animal procedures were approved by the Institutional Animal Care and Use Committee at the University of California, San Francisco (protocol AN174991). [Optional: state which institutional or national animal care and use guidelines were followed—e.g. the NIH Guide for the Care and Use of Laboratory Animals. This is a “should” under IOP policy and is not stated in Comrie et al.; confirm the standard with the Frank lab before adding.]

Acknowledgements

This work was supported by the National Institutes of Health (R01MH136875 to L.M.F.), the Simons Collaboration on the Global Brain (521921 and 542981 to L.M.F.), and the Howard Hughes Medical Institute (L.M.F.). A.E.C. was supported by the National Science Foundation Graduate Research Fellowship Program (1650113), the National Institutes of Health (F31MH124366), and the UCSF

Jonas Cohler Discovery Fellowship. [TODO: add funding for U.T. Eden and any support for S. Zeng and E.L. Denovellis, and state whether the funders had any role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.]

The authors declare that they have no competing interests.

Author contributions

Conceptualization: UTE, SZ.

Methodology: UTE, SZ.

Software: SZ, ELD.

Formal analysis: SZ, ELD.

Investigation: SZ, UTE, ELD.

Data curation: AEC.

Writing – original draft: SZ, UTE, ELD.

Writing – review & editing: SZ, AEC, LMF, UTE, ELD.

Visualization: SZ, ELD.

Supervision: UTE.

Funding acquisition: UTE, LMF.

References

- [1] Jones L M, Fontanini A, Sadacca B F, Miller P and Katz D B 2007 *Proc. Natl. Acad. Sci. U.S.A.* **104** 18772–18777 ISSN 0027-8424, 1091-6490
- [2] Denovellis E L, Gillespie A K, Coulter M E, Sosa M, Chung J E, Eden U T and Frank L M 2021 *eLife* **10** e64505 ISSN 2050-084X
- [3] Vinograd A, Nair A, Kim J H, Linderman S W and Anderson D J 2024 *Nature* **634** 910–918 ISSN 0028-0836, 1476-4687

- [4] Newman K B, Buckland S T, Morgan B J T, King R, Borchers D L, Cole D J, Besbeas P, Gimenez O and Thomas L 2014 *Modelling Population Dynamics: Model Formulation, Fitting and Assessment Using State-Space Methods* Methods in Statistical Ecology (New York, NY: Springer New York) ISBN 978-1-4939-0976-6 978-1-4939-0977-3
- [5] Deng X, Liu D F, Kay K, Frank L M and Eden U T 2015 *Neural Computation* **27** 1438–1460 ISSN 0899-7667, 1530-888X
- [6] Tao L, Weber K E, Arai K and Eden U T 2018 *J Comput Neurosci* **45** 147–162 ISSN 0929-5313, 1573-6873
- [7] Zeng S and Eden U T 2026 *IEEE Trans. Biomed. Eng.* **73** 153–167 ISSN 0018-9294, 1558-2531
- [8] Stone M 1974 *Journal of the Royal Statistical Society Series B: Statistical Methodology* **36** 111–133 ISSN 1369-7412, 1467-9868
- [9] Hastie T, Tibshirani R and Friedman J H 2009 *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* vol 2 (Springer) ISBN 978-0-387-84858-7
- [10] Shen X and Zhuang J 2024 *IEEE Trans. Emerg. Top. Comput. Intell.* **8** 1728–1743 ISSN 2471-285X
- [11] Bayarri M J and Castellanos M E 2007 *Statist. Sci.* **22** ISSN 0883-4237
- [12] Moran G E, Blei D M and Ranganath R 2024 *Journal of the Royal Statistical Society Series B: Statistical Methodology* **86** 194–214 ISSN 1369-7412, 1467-9868
- [13] Guttman I 1967 *Journal of the Royal Statistical Society: Series B (Methodological)* **29** 83–100 ISSN 0035-9246
- [14] Meng X L 1994 *The annals of statistics* **22** 1142–1160 ISSN 0090-5364
- [15] Gelman A, Meng X L and Stern H 1996 *Statistica sinica* 733–760 ISSN 1017-0405

- [16] Kullback S and Leibler R A 1951 *The annals of mathematical statistics* **22** 79–86 ISSN 0003-4851
- [17] Comrie A E, Monroe E J, Kahn A E, Denovellis E L, Joshi A, Guidera J A, Krausz T A, Berke J D, Daw N D and Frank L M 2024 *bioRxiv* 2024.09.23.613567
- [18] Chung J E, Magland J F, Barnett A H, Tolosa V M, Tooker A C, Lee K Y, Shah K G, Felix S H, Frank L M and Greengard L F 2017 *Neuron* **95** 1381–1394.e6 ISSN 08966273
- [19] Wilson M and McNaughton B 1994 *Science* **265** 676–679 ISSN 0036-8075, 1095-9203
- [20] Karlsson M P and Frank L M 2009 *Nature Neuroscience* **12** 913–918 ISSN 1097-6256, 1546-1726
- [21] Bhattacharyya A, Gayen S, Meel K S, Myrasiotis D, Pavan A and Vinodchandran N 2024 Total variation distance meets probabilistic inference *Proceedings of the 41st International Conference on Machine Learning* pp 3776–3794
- [22] Crack T F 2018 *SSRN Journal* ISSN 1556-5068
- [23] Deng X, Liu D F, Karlsson M P, Frank L M and Eden U T 2016 *Journal of Neurophysiology* **116** 2221–2235 ISSN 0022-3077, 1522-1598